TALHA ZAIN

Applied AI Engineer | LLM Systems · Agentic AI · Production ML
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PROFILE

Applied AI Engineer with professional experience since January 2023 building AI products and backend platforms across multimodal generation, conversational and voice AI, clinical systems, and geospatial recommendation. Hands-on owner of Python/FastAPI services and LangGraph-based agent orchestration, with work spanning retrieval evaluation, asynchronous jobs, multi-tenant access controls, and production performance optimization. Systems include a generative AI platform serving 60K+ monthly active users and conversational deployments processing 12K+ chats daily.

TECHNICAL EXPERTISE

LLM & Agents: LangGraph, LangChain, tool/function calling, multi-agent orchestration, agent state, human-in-the-loop, structured outputs, MCP

Retrieval & Evaluation: RAG, graph-based retrieval, hybrid search, BM25, RRF, reranking, Qdrant, pgvector, FAISS, RAGAS, LangSmith

Backend & Data: Python, FastAPI, Django, Flask, REST APIs, AsyncIO, PostgreSQL, Redis, MongoDB, SQL, queues, webhooks, indexing, connection pooling

ML, Cloud & Delivery: PyTorch, TensorFlow, scikit-learn, Hugging Face, Docker, Kubernetes, AWS, GCP, Azure, CI/CD, MLflow

PROFESSIONAL EXPERIENCE

BoolMind — ML Engineer, Applied / Generative AI | Lahore | Jul 2025 – Present

- Architected a semantic outdoor-advertising recommendation pipeline across 160,000 UAE points of interest (POIs); translated campaign requirements into structured geographic intent and retrieved relevant POIs within a 500 m radius of inventory screens.
- Developed a weighted recommendation approach incorporating footfall (30%), traffic (30%), POI density (20%), and proximity (20%) across 290+ UAE categories; achieved 99% stakeholder acceptance of recommendations and search response times under 3 seconds.
- Built Python/FastAPI AI services and data integrations with retries, caching, validation, and idempotent update patterns; optimized request paths through database indexing, connection pooling, and asynchronous processing.
- Owned the Python/FastAPI backend for AI Compare Hub, a multimodal generation platform with 60K+ monthly active users and 93% generation success; optimized database indexing, connection pooling, and threaded processing.
- Engineered production MedicAI agent orchestration with LangGraph and graph-based RAG for clinical tasks; measured 98% retrieval hit@k against labeled evaluation data.
- Led technical decisions, deployments, CI/CD, code reviews, and production issue resolution across AI services.

IT Genics — Associate AI Engineer | Lahore | Jan 2024 – Jun 2025

- Developed a no-code conversational AI platform deployed across 45 organizations (27 using both chat and voice agents; 18 using chat-only agents), processing 12,000+ chats per day.
- Integrated Llama-based agents, Twilio-powered voice interactions, external APIs, and business actions for customer support, bookings, and information retrieval.
- Achieved a reported 99% task-completion success rate; the company reported a 21% revenue increase associated with the platform. Contributed to application delivery, APIs, and product collaboration.

IT Solutions World Wide — Junior AI Engineer | Islamabad | Mar 2023 – Dec 2023

- Developed NLP applications using GPT-3, Llama, BERT, and Falcon, incorporating prompt-based techniques, transfer learning, and fine-tuning for domain-oriented language tasks.
- Implemented reusable model-serving APIs and ML pipelines using Python, PyTorch, and TensorFlow.

Adveneurs — Junior Data Analyst | Jan 2023 – Jun 2023

SELECTED AI SYSTEMS

MediaX — Semantic & Geospatial OOH Recommendation | map.mediaxnetwork.com

- Built an AI-assisted campaign-planning and inventory-ranking pipeline using 160K UAE POIs, 500 m screen-radius searches, and 290+ POI categories; achieved 99% stakeholder acceptance of recommendations with responses under 3 seconds.

MoverGPT — No-Code Chat & Voice Agent Platform | IT Genics

- Built configurable conversational and voice assistants integrating Llama, Twilio, API actions, and customer workflows; deployed across 45 organizations (27 chat + voice; 18 chat-only) with 12K+ daily chats and reported 99% task-completion success.

Recruiter AI — Recruitment & ATS Platform | github.com/talhazaindev/recruiter-ai

- Developed a recruitment platform for job-description and resume ingestion, document parsing, queued matching, candidate shortlisting, and human review; implemented with FastAPI, MongoDB, Redis workers, MinIO, and a React/TypeScript interface.

AI Compare Hub — Multimodal Model Platform | ai-compare-hub.com

- Integrated 30+ image, 20+ video, and 5+ audio models with asynchronous provider jobs, status tracking, retries, and side-by-side comparison of up to four models per prompt.

MedicAI — Multi-Agent Clinical Platform | Production

- Orchestrated clinical agents with LangGraph, graph-based retrieval, role-based access for doctors, patients, and hospital staff, and human approval gates.

EDUCATION & CERTIFICATIONS

B.S. Computer Science | National University of Computer & Emerging Sciences (FAST-NUCES), Islamabad | 2019–2023

Microsoft Azure Fundamentals (AZ-900) | IBM Machine Learning with Python | IBM Databases & SQL for Data Science